

```
#linear regression  
#logistic regression
```

```
#supervised regression (labelled)  
#unsupervised regression (unlabelled)
```

Start coding or [generate](#) with AI.

```
import pandas as pd  
df=pd.read_csv('/content/Student_Performance.csv')  
df.head()
```



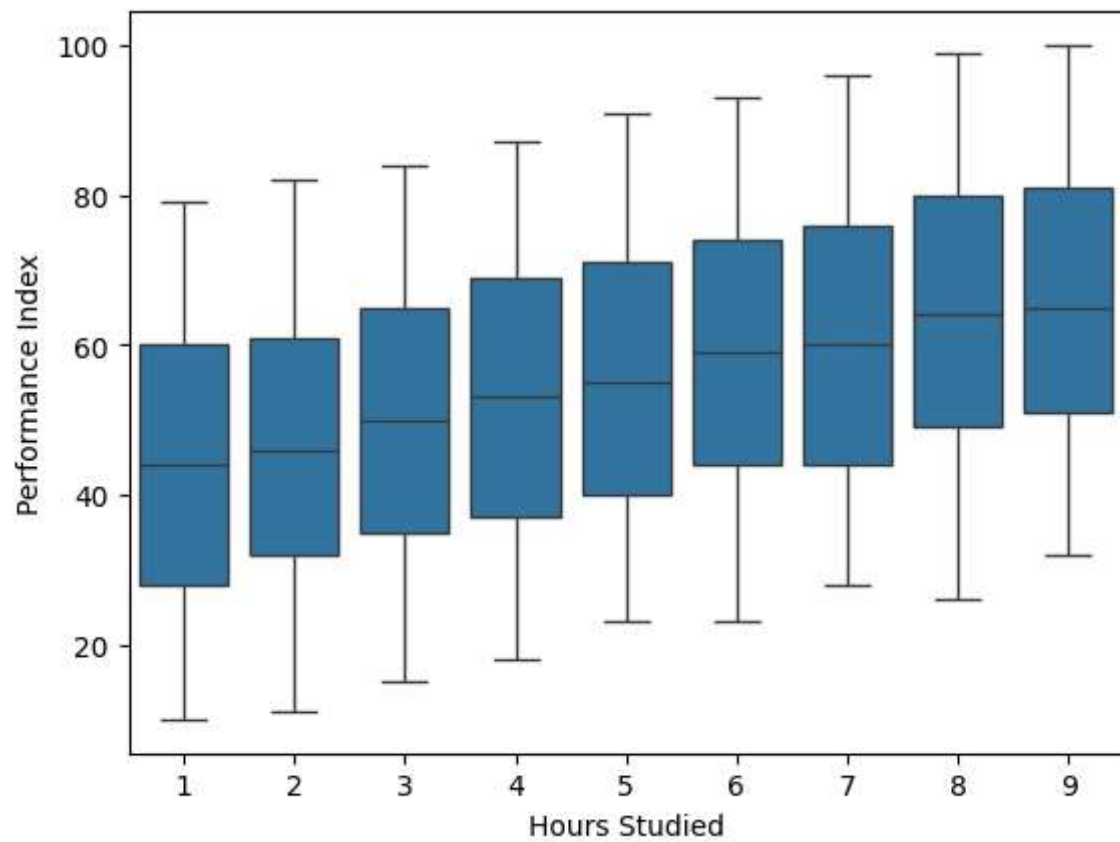
	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
0	7	99	Yes	9	1	91.0
1	4	82	No	4	2	65.0
2	8	51	Yes	7	2	45.0
3	5	52	Yes	5	2	36.0
4	7	75	No	8	5	66.0

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
import seaborn as sns  
  
sns.boxplot(x='Hours Studied', y='Performance Index', data=df)
```

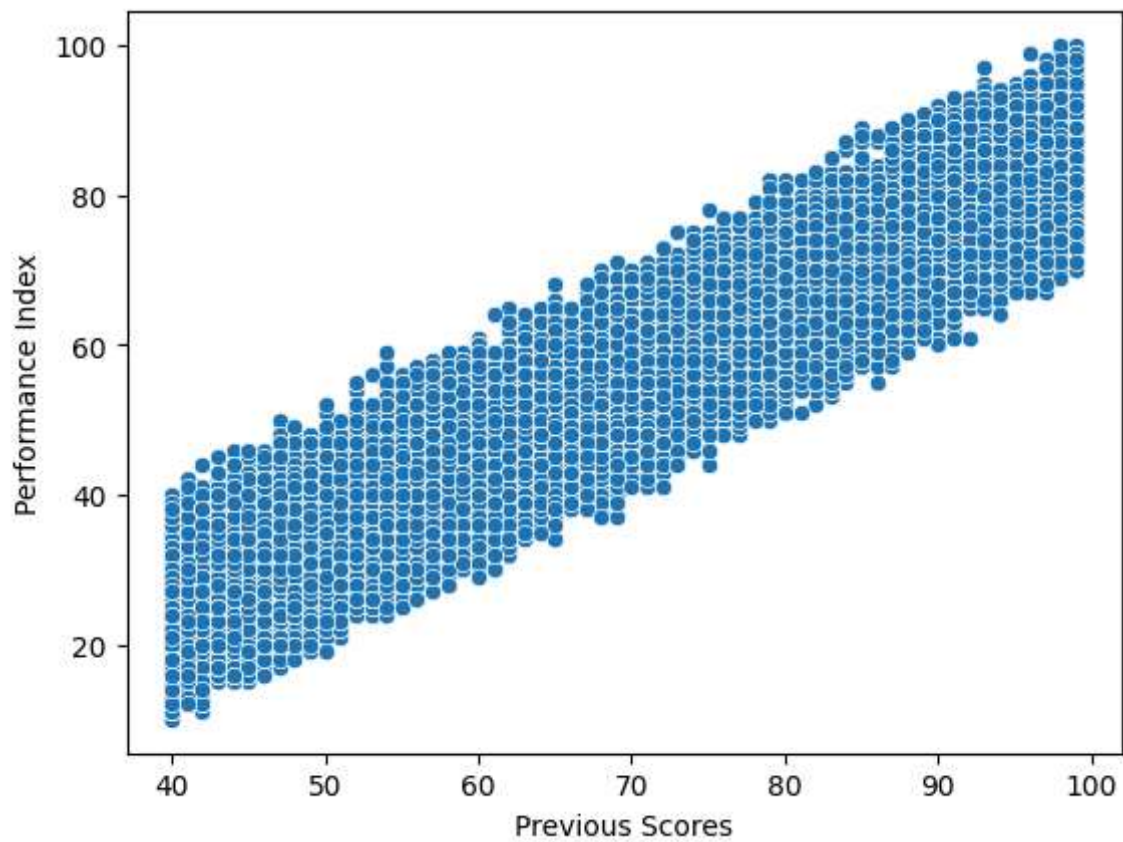
<Axes: xlabel='Hours Studied', ylabel='Performance Index'>



Double-click (or enter) to edit

```
sns.scatterplot(x='Previous Scores',y='Performance Index',data=df)
```

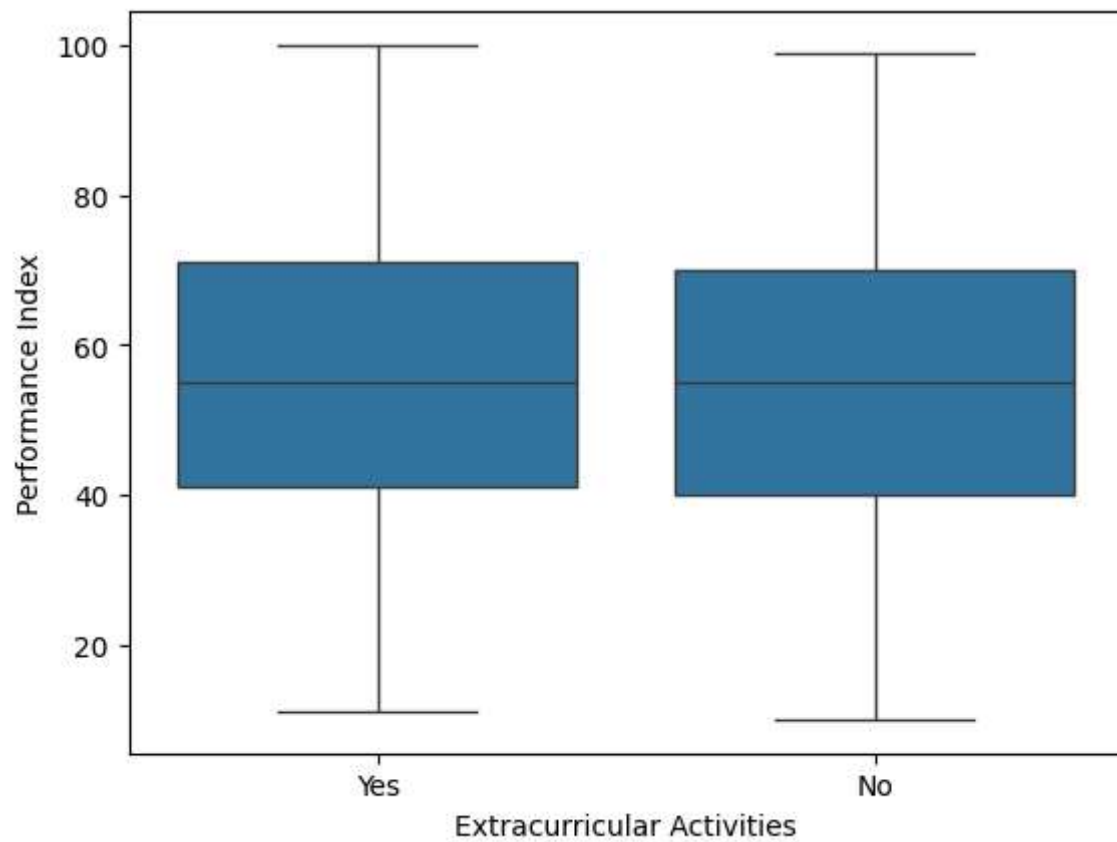
<Axes: xlabel='Previous Scores', ylabel='Performance Index'>



Double-click (or enter) to edit

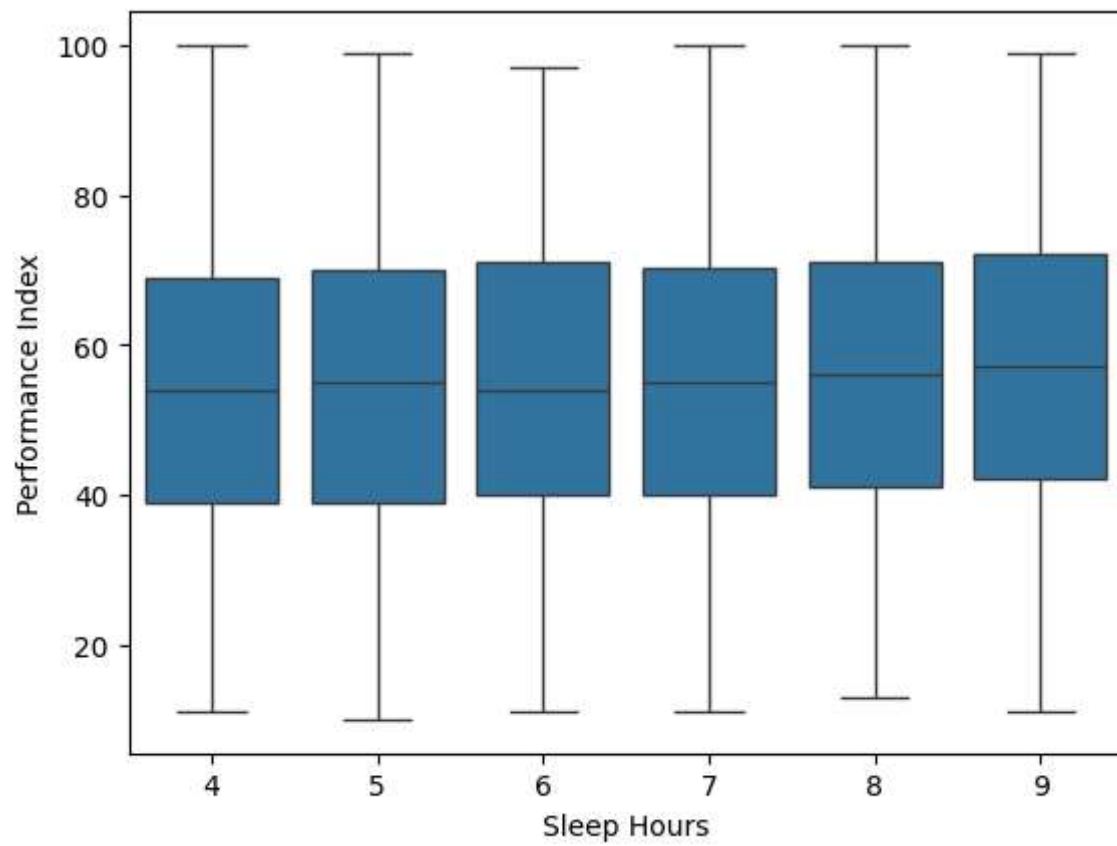
```
sns.boxplot(x='Extracurricular Activities',y='Performance Index',data=df)
```

<Axes: xlabel='Extracurricular Activities', ylabel='Performance Index'>



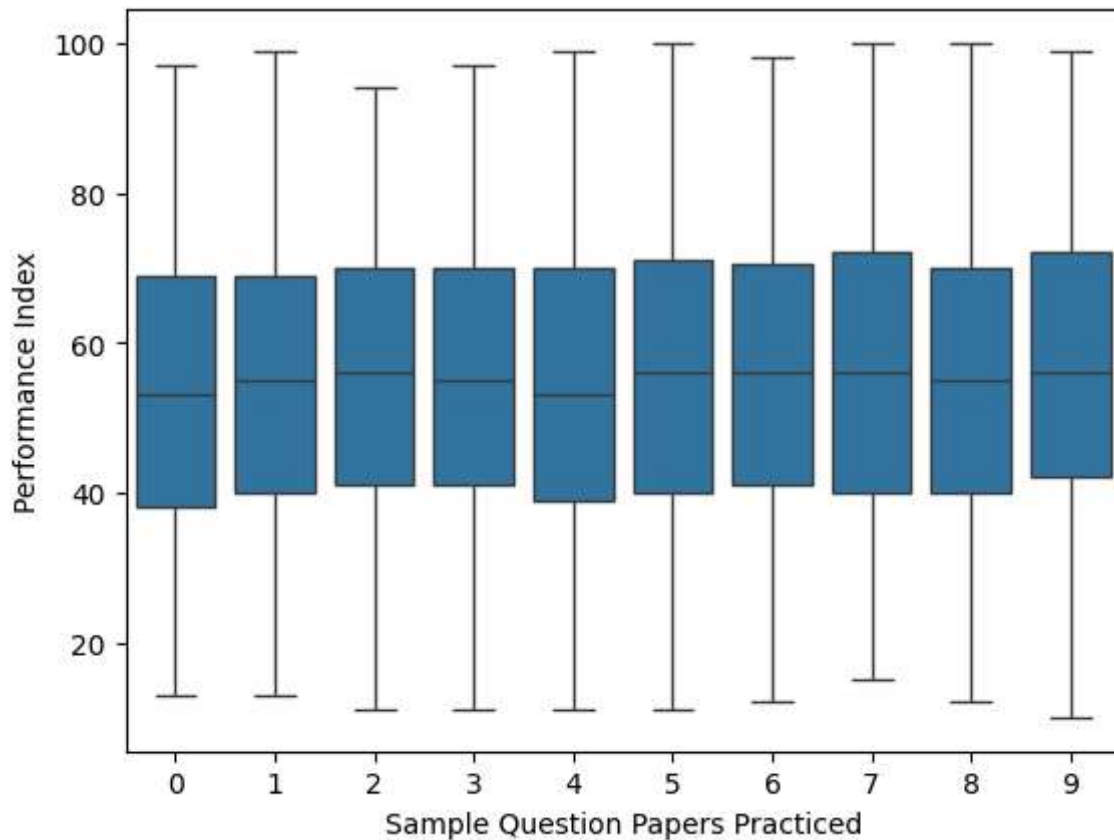
```
sns.boxplot(x='Sleep Hours',y='Performance Index',data=df)
```

<Axes: xlabel='Sleep Hours', ylabel='Performance Index'>



```
sns.boxplot(x='Sample Question Papers Practiced',y='Performance Index',data=df)
```

<Axes: xlabel='Sample Question Papers Practiced', ylabel='Performance Index'>



#converting columns categories into a no.(for logistic reg)

```
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
df['Extracurricular Activities']=le.fit_transform(df['Extracurricular Activities'])
df.head()
```

	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
0	7	99	1	9	1	91.0
1	4	82	0	4	2	65.0
2	8	51	1	7	2	45.0
3	5	52	1	5	2	36.0
4	7	75	0	8	5	66.0

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
X=df.drop('Performance Index',axis=1)
```

`y=ar[ Performance Index ]`

